

FujiNet cable wiring: SD daughter board & host SIO

Two cables to build. One is trivially straight-through. The other is not — the SIO header drops a pin partway along, so half its conductors land on a different number at the far end, and that far end is soldered directly to filter-cap pads on the 130XE motherboard rather than to a connector.

FN32ROV-XEBook rev 1.0 Board files verified 2026-08-01
Host: Atari 130XE C070065 / C070067

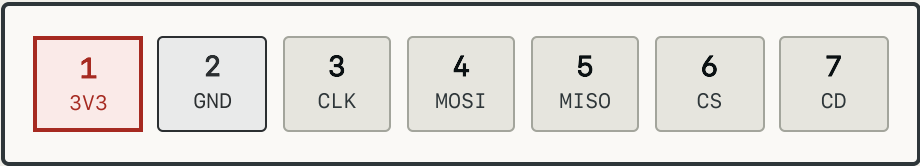
CABLE 1 OF 2 · LOW RISK

SD daughter board → main board

7-conductor JST PH cable, **2.00 mm pitch**, straight through 1:1. Both ends use the same horizontal connector (`S7B-PH-K`), so an off-the-shelf 7-pin JST PH pigtail works as-is. Pin 1 to pin 1, no crossover anywhere.

THE ONLY WAY TO GET THIS WRONG

Reversing the cable end-for-end. Pin 1 carries 3V3 and pin 2 is ground; flipped, 3V3 lands on the card-detect line and the SD socket sees power on its switch contact. Confirm pin 1 at both ends against the board silkscreen before first power-up.



Identical at both ends — J_SD1 on the main board, J1 on the SD board. Square pad marks pin 1.

CONDUCTOR MAP — 1:1, NO CROSSOVER

PIN	SIGNAL	MAIN BOARD J_SD1	SD BOARD DESTINATION
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PIN	SIGNAL	MAIN BOARD J_SD1	SD BOARD DESTINATION
1	3V3	3V3 rail	SD1 pin 4 (VDD), C1
2	GND	Ground	SD1 pins 3 & 6 (VSS), shield tabs, CP, C1
3	SPI_CLK	ESP32 IO18	SD1 pin 5 (CLK)
4	SPI_MOSI	ESP32 IO23	SD1 pin 2 (CMD)
5	SPI_MISO	ESP32 IO19	SD1 pin 7 (DAT0)
6	SPI_CS	ESP32 IO5	SD1 pin 1 (CD/DAT3)
7	SD_CD	ESP32 IO15	SD1 card-detect switch (SW)

TWO BEHAVIOURS TO EXPECT

One card at a time. This socket sits in parallel with the main board's onboard microSD on the same SPI bus, with no multiplexer. Two cards inserted at once puts two devices on one bus.

DAT1/DAT2 float on this socket. The 7-pin cable has no line for them, and the pull-ups (R27/R28) reach only the onboard socket. SPI mode normally tolerates this. If cards enumerate unreliably, add 10 kΩ from SD1 pins 8 and 9 to 3V3 — the contact row is 2.5 mm pitch and workable by hand.

CABLE 2 OF 2 · GETS IT WRONG AND SOMETHING DIES

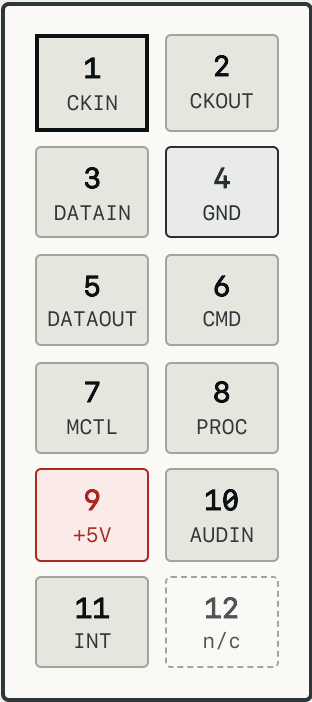
J_SIO1 → 130XE SIO port

12-way IDC header on the FujiNet, 11 conductors used, landing on the Atari's 13-pin SIO signals. The original design carried two full SIO connectors; this board replaces them with one low-profile header hardwired to the host.

IDC PIN NUMBERS ARE NOT SIO PIN NUMBERS

SIO's second ground (pin 6) was dropped, so **everything from IDC pin 6 upward shifts by one**. IDC 1–5 match SIO 1–5; after that IDC 6 → SIO 7, IDC 7 → SIO 8, and so on. Wire it straight through and **SIO pin 10 (+5 V) lands on a logic input**.

There are no per-pin labels silkscreened on the board — only the reference designator. This table is the authority.



J_SIO1 as it physically sits — 2 columns × 6 rows, 2.54 mm pitch. Pin 1 is the **square pad**; odd pins run down the left column, even down the right. Ribbon conductor 1 goes to pin 1 and the cable alternates columns as it descends.

 power ground signal not connected

CONDUCTOR MAP — NOTE THE SHIFT FROM IDC PIN 6

IDC PIN	SIGNAL	SIO PIN	DIRECTION / NOTES
1	SIO_CKIN	1	Out from FujiNet (open collector, U4)
2	SIO_CKOUT	2	In to FujiNet

IDC PIN	SIGNAL	SIO PIN	DIRECTION / NOTES
3	SIO_DATAIN	3	Out from FujiNet; 4.7 kΩ pull-up to SIO +5 V
4	GND	4	The only ground conductor — see below
5	SIO_DATAOUT	5	In to FujiNet
6	SIO_CMD	7 ←	In to FujiNet; shift starts here
7	SIO_MCTL	8 ←	In to FujiNet; 2 kΩ pull-down
8	SIO_PROC	9 ←	Out from FujiNet
9	SIO_5V	10 ←	+5 V in — board supply, 47 μF bulk, OR-ing diode
10	SIO_AUDIN	11 ←	Out from FujiNet (ESP32 DAC)
11	SIO_INT	13 ←	Out from FujiNet
12	—	—	Free pad. Bridge to pin 4 — see below

Deliberately not carried: SIO pin 6 (second ground) and SIO pin 12 (+12 V, which FujiNet doesn't use and the XE doesn't supply).

ADD A SECOND GROUND RETURN WHILE YOU BUILD THIS

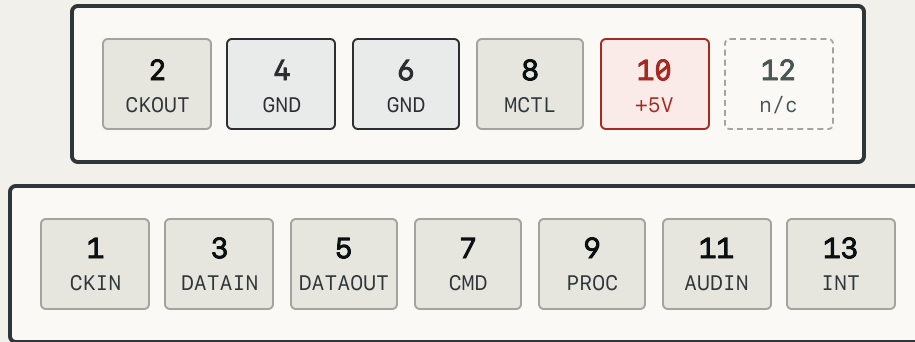
With SIO pin 6 dropped, the board's entire supply current returns through IDC pin 4 alone, on a single 0.2 mm trace. **IDC pin 12 is a free floating pad** — bridge it to pin 4 and run it to SIO pin 6, giving a second return path. Doing it in the cable costs one extra conductor now; the alternative is waiting for the next board spin.

FAR END · 130XE MOTHERBOARD C070065 / C070067

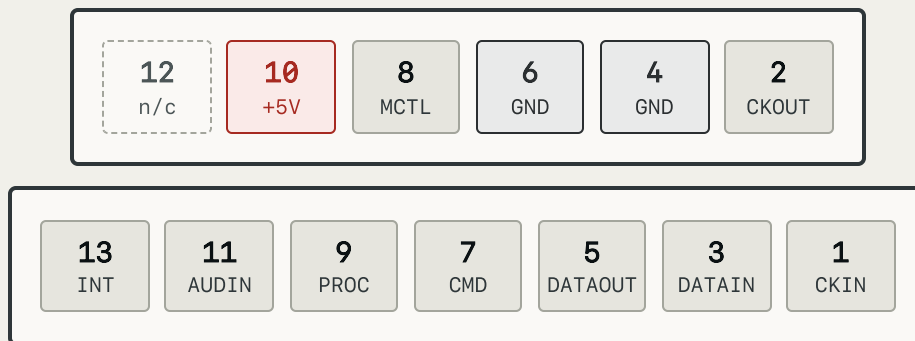
Where to land the wires

You're not wiring into the SIO jack itself — you're tapping the filter capacitors beside it. That works cleanly here because **every SIO signal FujiNet needs has its own dedicated ceramic cap at J1**, with one leg on the signal and the other on ground. Ten signals, ten caps, all in two short rows.

FRONT – LOOKING INTO THE JACK FROM OUTSIDE



BACK – FACING THE FRONT OF THE MOTHERBOARD · THIS IS THE ONE YOU SOLDER TO



J1 SIO jack, both views. Evens on the upper row, odds on the lower. Front view per Figure 1-12 of the field service manual; back view is its mirror, cross-checked against *SIO Wiring.psd*.

THE TWO VIEWS ARE MIRROR IMAGES – DON'T MIX THEM UP

From the front, the upper row runs **2 4 6 8 10 12** left to right. From the back it runs **12 10 8 6 4 2**. Everything reverses, including which end +5 V sits at. Working from the front-view drawing while looking at the back of the board is the single most common way this job goes wrong.

TAP POINTS – C070065 / C070067 REFERENCE DESIGNATORS

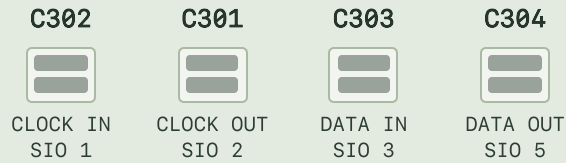
IDC PIN	SIGNAL	SIO PIN	SOLDER TO	VALUE
1	SIO_CKIN	1	C302	27 pF
2	SIO_CKOUT	2	C301	27 pF

IDC PIN	SIGNAL	SIO PIN	SOLDER TO	VALUE
3	SIO_DATAIN	3	C303	27 pF
4	GND	4	Ground leg of any cap below	—
5	SIO_DATAOUT	5	C304	27 pF
6	SIO_CMD	7 ←	C308	1 nF
7	SIO_MCTL	8 ←	C306	1 nF
8	SIO_PROC	9 ←	C307	1 nF
9	SIO_5V	10 ←	C310	1 nF
10	SIO_AUDIN	11 ←	C305	1 nF
11	SIO_INT	13 ←	C309	1 nF
12	GND (2nd return)	6	Ground leg of any cap	—

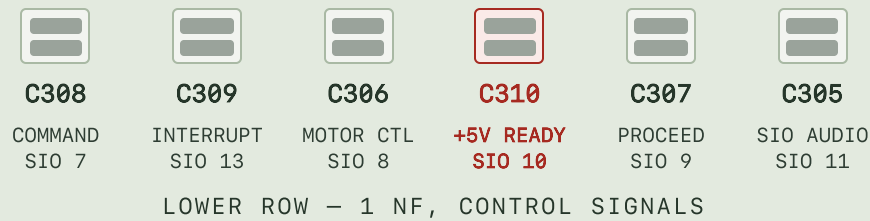
Where they physically sit

Plan view matching the photograph: SIO connector at the left with its pins pointing right, the six control-signal caps in a row below it, the four clock/data caps in a group above-right. **The silkscreen on the board confirms the scrambled order.**

UPPER GROUP – 27 PF, CLOCK & DATA



J1 SIO – 13 pins, body to the left, pins to the right



+5 V IS THE FOURTH PAD FROM THE LEFT

In the lower row, **C310 carries +5 V and sits between two logic signals** — MOTOR CONTROL to its left, PROCEED to its right. It looks like every other pad on the row. Miscount by one in either direction and +5 V goes onto a POKEY line.

THE PADS LOOK UNPOPULATED ON YOUR BOARD

In the photo both groups appear to be bare tinned pads with no capacitors fitted, which is why they're free for you to use. Worth confirming, because it changes the method: with no cap body bridging the pair, **you can't identify the signal leg by following the component** — you have to buzz each pad to its SIO jack pin. The signal pad still connects through its trace whether or not the cap is there.

THE DESIGNATORS ARE NOT IN SIGNAL ORDER

C305 is *audio*, C308 is *command*, C310 is *+5 V*. The numbering runs in physical board order, not signal order, so reading the row left to right and assuming C305→C310 matches the pin sequence will put **+5 V onto the audio line**. Use the table, not the sequence.

PICK THE CORRECT LEG OF EACH CAP

Every one of these caps has **one leg on the signal and one leg on ground**. Solder to the signal leg. Buzz each pad against its SIO jack pin before committing — a cap that reads continuity to chassis ground is the wrong leg, and it's a dead short of that signal to ground.

The upside: this also gives you ground everywhere. Any cap's ground leg serves for IDC pin 4, and a second one for the pin 12 return.

WHY THE CAPS AND NOT THE RESISTORS

R128–R131 (100 Ω) are series damping resistors on the four clock/data lines, sitting *between POKEY and the connector*. The 27 pF caps are downstream of them, on the connector side — the same side external SIO peripherals see. That's the correct side to tap, and it's why C301–C304 are the landing points rather than the resistor pads. The other six signals have no series resistor at all, only their 1 nF cap.

Leave every capacitor in place. They're RF filtering on a machine that had to pass emissions testing; removing them to free a pad invites trouble on the audio and clock lines especially.

CONFIRM THE BOARD REVISION FIRST

These designators are from the **130XE CO70065 / CO70067** schematic, Rev 5. The SECAM CO70052, the 65XE CO70025 and CA20097, and the CA200519 boards are laid out differently and **their designators do not carry over**. Check the silkscreen near the RF shield before you trust this table — then verify by meter regardless.

Verification before soldering

1. Power off and unplug. Let the PSU sit a minute.
2. Find SIO jack pin 4 and confirm continuity to chassis ground. That anchors your orientation and tells you which way the numbering runs on your board.
3. For each signal, buzz from its SIO jack pin to the cap you intend to use. Continuity confirms both the designator and the correct leg.
4. Mark the confirmed leg of each cap before you start — flux and solder look identical at 2 a.m.
5. After soldering, power on and confirm the wire destined for IDC pin 9 reads **+5 V** and nothing else does.

Build and test

Continuity checks

- Every conductor end to end, one at a time, against the tables above.
- No shorts between physically adjacent pins — the failure mode IDC crimping actually produces.
- SIO cable: confirm IDC 9 reaches SIO 10 and nothing else. This is the +5 V conductor.
- SIO cable: confirm IDC 12 ties to IDC 4 if you added the second ground.
- SD cable: pin 1 to pin 1, both ends, before it goes anywhere near power.

Bring-up order

1. Fit a switch or jumper across J_PWR1 — the regulator's enable is held low, so an open jumper means a dead board.
2. Fit the U.FL antenna. The module is a WROVER-**IE**; there is no PCB antenna.
3. Power over USB-C first. Confirm the 3V3 rail, then that it enumerates as a CP2102N.
4. Flash firmware over USB.
5. Only then connect the SIO cable to the 130XE.

POWER BUDGET WORTH KNOWING

The ESP32 peaks in the hundreds of milliamps when WiFi transmits, and on SIO that current comes from the Atari's +5 V rail through one conductor. The OR-ing diodes (D7/D8) let USB-C carry the load instead — worth using while bench-testing rather than leaning on the 130XE's supply.

Pin assignments extracted directly from FN32ROV-XEBook-KiCad.kicad_pcb and FN32ROV-XEBook-SD-KiCad.kicad_pcb, not transcribed from documentation.

IDC geometry read from IDC-Header_2x06_P2.54mm_Latch_Horizontal.kicad_mod.

130XE tap points traced from Figure 2-1, Schematic Atari 130XE 16 X 4164 Assembly C070065/C070067 Rev 5; jack orientation from Figure 1-12, both in the 130XE Field Service Manual. Verify by meter before soldering — board revisions differ.